Modified Checklist for Autism in Toddlers

Name	Modified Checklist for Autism in Toddlers R/F (MCHAT)
Table Name	`mh_ch_mchat`
Construct	Autism
Study Visits	V07
Type	Remote Questionnaire (Parent on Child; 5-10 min)
Quality Control	• All item-level responses and scores were electronically captured and scored within the HBCD database. • Data was reviewed to verify that failed item level responses generated a follow-up response. • Completion rates were monitored for the overall consortium and at the site level to address any concerns.

Responsible Use Warning

The HBCD dataset includes many variables that may be important for sound and comprehensive analysis. The inclusion of additional variables will depend on the research question(s) and methodological approach. Users are encouraged to take time to explore the full range of available variables—especially those that may serve as controls, contextual indicators, confounders, mechanisms, or modifiers—to ensure thoughtful and well-supported analytic decisions. Other important considerations may include developmental functioning, broader family supports, and early adverse and protective exposures.

Data Warning

V07 and V07a administration

In the initial stage of data collection, the M-CHAT-R/F was administered at the V07a visit before it was removed from the protocol. It is now only included at the V07 visit.

A small number of participants (N=28) were administered the M-CHAT-R/F twice (at both V07a and V07). In those cases, only the M-CHAT-R/F data collected at V07 are included in the data release.

Four participants only completed the M-CHAT-R/F at V07a. These data wil
the V07 data in a later release.

Five participants completed the M-CHAT-R/F at V07a but at the time of the data release had not yet completed the M-CHAT-R/F at V07. The V07a data are not included in the current data release. In a later release, only the V07 data will be included in the data set and the M-CHAT-R/F at V07a will not be included in the data release.

See Known Issues & Pending Updates for issues that may affect data use.

Instrument Details

The M-CHAT-R/F (Robins et al., 2014) is a two-stage parent-report screening tool to assess the likelihood for autism. There are 20 questions. If a child fails an item, the caregiver is later interviewed and presented with additional questions for clarification. In the web-based, electronic version of the M-CHAT-R/F (Attar et al., 2023), the caregiver is instructed to follow instructions as presented onscreen. The initial responses and the follow-up responses for each question are included in the data release.

The M-CHATR/F is available as a free download for clinical, research, and educational purposes (available in English and Spanish). See Resources below.

HBCD Modifications

Modifications Associated with Web-Based Administration

If a child fails an item the caregiver is immediately presented with the additional questions (rather than administering all items in the screener form prior to moving on to follow-up interview questions. This modification was applied for ease of completion in web-based format to group related questions,

This modification means that some participants received follow-up questions that they would not have in the validated paper-administered version (i.e., screener scores of 0-2).

Scoring Procedures

Scoring procedures are included for each question to determine if the responses to the additional questions result in a pass or a fail and whether follow-up interview will be administered.

For all items except 2, 5, and 12, the response “NO” indicates elevated likelihood of autism; for items 2, 5, and 12, “YES” indicates elevated likelihood of autism. The total of these responses is used to determine if the results are associated with the criterion for High Risk (follow-up score of 2 or higher) or Low Risk (score of 0 or 1).

The follow-up questions and scoring procedures for the web-based, electronic version of the M-CHAT-R/F were provided to HBCD by one of the t
(Diana Robins, Ph.D.).

3.0-readmes

Resources

- Web-based, electronic version of the M-CHAT-R/F
- Official M-CHAT: The Modified Checklist for Autism in Toddlers, Revised with Follow-Up

References

Robins, D.L., Casagrande, K., Barton, M.L., Chen, C., Dumont-Mathieu, T., & Fein, D. (2014). Validation of the Modified Checklist for Autism in Toddlers-Revised with Follow-Up (M- CHAT-R/F). *Pediatrics*, 133 (1), 37-45. PMID:24366990. PMCID: PMC3876182.

Attar SM, Bradstreet LE, Ramsey RK, Kelly K, Robins DL. Validation of the Electronic Modified Checklist for Autism in Toddlers, Revised with Follow-Up: A Nonrandomized Controlled Trial. J Pediatr. 2023 Feb 2;262:113343. doi: 10.1016/j.jpeds.2022.11.044. Epub 2023 Feb 2. PMID: 36736890; PMCID: PMC10390646.